

PS/2 keyboard + mouse → Tang Nano 20K

Breadboard interface for the P8X graphics card, built on the **TXS0102** — one 2-channel translator per port.

WHY SHIFT

Nano GPIO are 3.3 V and *not* 5 V-tolerant; PS/2 is a 5 V bus. Direct wiring damages the FPGA.

WHY TXS0102

Auto-direction, open-drain-native — *no direction pin*, exactly what PS/2's directionless CLK/DATA need.

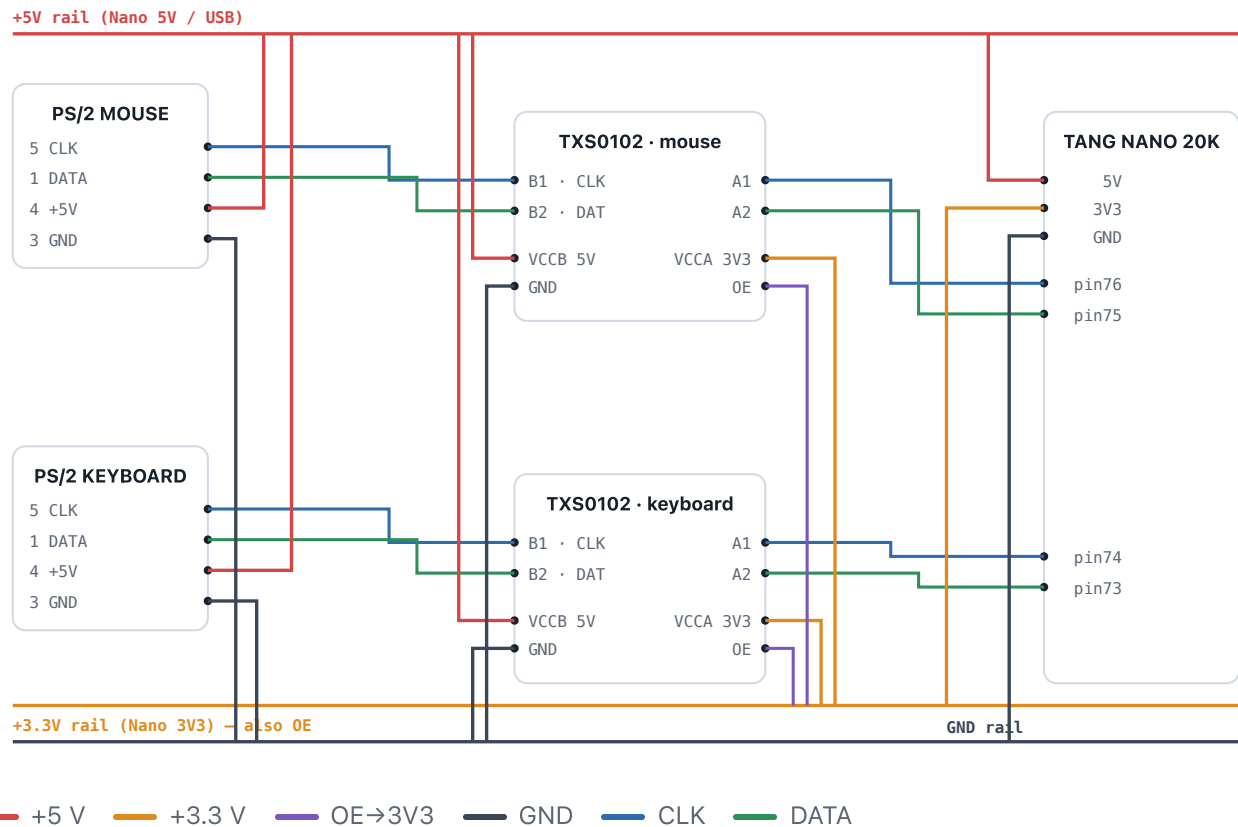
ONE CHIP PER PORT

2 channels each: CLK + DATA.
Mouse → one TXS0102,
keyboard → another.

FEWER PARTS

Internal ~10k pull-ups on both sides → *no external signal pull-ups*. Just power + OE + decoupling.

Connection diagram



No external pull-ups on the signal lines. The TXS0102 has ~10 kΩ pull-ups to VCCA and VCCB built into each channel — that is the PS/2 bus pull-up. Adding your own would fight the chip's edge-rate one-shot; leave them off.

Wiring table — the authoritative reference

PS/2	PIN	NET	TXS0102	→ B SIDE	A SIDE →	NANO
Mouse	5	CLK	#1 (mouse)	B1	A1	pin76
Mouse	1	DATA	#1 (mouse)	B2	A2	pin75
Keyboard	5	CLK	#2 (kbd)	B1	A1	pin74
Keyboard	1	DATA	#2 (kbd)	B2	A2	pin73
Both	4	+5V	both	VCCB	Nano 5V	
—	—	+3.3V	both	VCCA + 0E (both chips)		3V3
Both	3	GND	both	chip GND		GND

Nano pins 76/75/74/73 are examples — free on the current card build (used: 4, 15–20, 27–42, 48, 69, 70, 77, 81–84). Confirm they're broken out on your board and add them to `tangnano20k.cst` (below). Any four free GPIO work.

How the TXS0102 does it (one channel)

Each channel is a bidirectional pass gate with an **edge-rate accelerator** (a one-shot that briefly drives hard on a rising edge, then hands back to the internal pull-up) and **~10 kΩ pull-ups** to VCCA and VCCB. It **senses which side goes low first** and mirrors it to the other — no direction pin, which is why it suits PS/2, where either the host or the device can pull a line low at any time.

VCCA ≤ VCCB (3.3 ≤ 5 ✓) sets the two voltage domains. **OE high** enables all channels — tie it to VCCA so it comes up *after* both supplies. 0.1 μF on each VCC.

TXS0102 PIN	CONNECT TO
VCCA	3.3 V (the lower rail)
VCCB	5 V (the higher rail)
GND	GND
OE	3.3 V (enable)
A1 / A2	FPGA GPIO (3.3 V side)
B1 / B2	PS/2 CLK / DATA (5 V side)

Bill of materials

QTY	PART	NOTES
2	TXS0102 (breakout board)	one per PS/2 port; a breakout is far easier than the bare VSSOP on a breadboard
2	PS/2 (mini-DIN 6) female socket	socket, or cut a PS/2 extension cable and meter the pins
4	0.1 μF ceramic	one on VCCA and one on VCCB of each chip
2	0.1 μF ceramic (optional)	across each socket's +5V↔GND
—	breadboard + ~12 jumpers	—
1	Tang Nano 20K	the P8X graphics card

No BSS138s, no discrete pull-up resistors — the two TXS0102s replace the 4-channel module and its pull-ups.

Build order

- Rails first.** Nano 5V → red rail, 3V3 → a second rail, GND → blue rail. Meter 5.0 V and 3.3 V *before* anything else.

2. **Each TXS0102:** VCCB→5 V, VCCA→3.3 V, GND→GND, **OE→3.3 V**, 0.1 μF on each VCC. (OE low = all channels Hi-Z, so don't forget it.)
3. **Sockets.** pin 4→5 V, pin 3→GND, optional 0.1 μF across them. **Meter the pins** if you cut a cable — colours aren't standard.
4. **Signals.** Mouse CLK(5)→#1 B1, DATA(1)→#1 B2; keyboard CLK(5)→#2 B1, DATA(1)→#2 B2. Then #1 A1→pin76, A2→pin75; #2 A1→pin74, A2→pin73.
5. **Constrain + build.** Add the pins to tangnano20k.cst, wire them to the (future) PS/2 receiver in the RTL, rebuild the bitstream.

FPGA side — constraints & the open-drain rule

```
// tangnano20k.cst - four free pins (confirm against your board)
IO_LOC  "ms_clk"  76;    IO_PORT "ms_clk"  IO_TYPE=LVC MOS33 PULL_MODE=NONE;
IO_LOC  "ms_dat"  75;    IO_PORT "ms_dat"  IO_TYPE=LVC MOS33 PULL_MODE=NONE;
IO_LOC  "kb_clk"  74;    IO_PORT "kb_clk"  IO_TYPE=LVC MOS33 PULL_MODE=NONE;
IO_LOC  "kb_dat"  73;    IO_PORT "kb_dat"  IO_TYPE=LVC MOS33 PULL_MODE=NONE;
```

The RTL drives each line OPEN-DRAIN, never push-pull: drive 0 to pull low, else Hi-Z and read the pin — assign `clk = clk_oe ? 1'b0 : 1'bz`; . This matters more with the TXS0102 than with a MOSFET shifter: a hard push-pull HIGH from the FPGA confuses its auto-direction sensing. Leave the on-chip pull as NONE — the TXS0102's internal pull-up already holds the A side high.

Don't wire any PS/2 pin straight to the Nano, don't add external pull-ups on A/B, and don't power the devices from 3.3 V — a PS/2 keyboard/mouse needs 5 V.

P8X · PS/2 interface (TXS0102) for the Tang Nano 20K graphics card. Pairs with the emulator \$FF58 model and lib_ps2; the RTL receiver (a shift register clocked by the device CLK) is the next hardware step.